Post-Disaster Damage Assessment and Rescue Guidance

CSE499B – Section 13, Group 5, Update 1

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Weekly Update:

Our primary objective this week was to architect and initiate a highly controlled, parallel Ablation Study to improve our DisasterM3 baseline segmentation model. Currently, the model performance is plateaued at 0.46 mean Intersection over Union (mIoU), falling short of our rigorous target metrics. The core challenge hindering performance is an extreme class imbalance inherent in satellite imagery of post-disaster zones, where the vast majority of pixels are uninformative background. To break through this plateau and aggressively optimize our pipeline, we designed a strict 15-day ablation sprint. We divided the optimization problem into four distinct domains: data sampling, architectural enhancements, loss function engineering, and inference resolution. Each track independently modifies exactly one variable to precisely measure its isolated impact on the Intersection over Union (IoU) of minority classes. This rigorous scientific approach ensures we avoid confounding variables and can confidently identify statistically significant improvements before ultimately fusing the best-performing configurations into a single, unified Champion Model.

Work Done by Member A - Track A

1. **Dataset Audit and Pixel Distribution Analysis:**
   - Developed a programmatic audit script utilizing OpenCV to iteratively analyze every training image's ground truth mask, extracting exact pixel counts and image-level class presence statistics.
   - The analysis exposed a severe foundational class imbalance: approximately 93% of all pixels in the training set represent empty background, while the most critical 'Destroyed' building class comprises a mere 0.33% of the dataset.
   - Generated comprehensive pixel distribution histograms, definitively proving that standard uniform sampling resulted in batches devoid of destroyed structures, causing severe gradient starvation for the minority classes during backpropagation.
2. **Class-Balanced Sampling Pipeline Integration:**
   - Engineered a custom `WeightedRandomSampler` integrated directly into the PyTorch `DataLoader` to proactively and aggressively mitigate the identified data imbalances before the data reaches the model.
   - Pre-computed dynamic sample weights by evaluating mask compositions, forcing a 10× oversampling rate for images containing 'Destroyed' structures and a 5× rate for images with 'Damaged' structures.
   - Disabled the default shuffle mechanism in favor of this weighted sampler, successfully ensuring the network receives a consistent, robust gradient signal for rare minority classes in every single training batch without reducing the base batch size of 16 (thereby preserving BatchNorm stability).

Work Done by Member B - Track B

1. **Baseline Validation and U-Net++ Implementation:**
   - Established a rigorously controlled baseline environment utilizing the `segmentation_models_pytorch` (SMP) API. Configured and trained a standard U-Net with a ResNet34 encoder (pre-trained on ImageNet) for 40 epochs to validate the 0.46 mIoU starting point.
   - Transitioned to implementing the advanced U-Net++ architecture. This model introduces nested dense skip connections that better bridge the semantic gap between the encoder's shallow, high-resolution feature maps and the decoder's deep, semantic maps, directly enhancing boundary delineation for complex building shapes.
2. **Attention Mechanism Integration (scSE) and ASPP Exploration:**
   - Conducted an isolated ablation experiment on the U-Net++ architecture by injecting Spatial and Channel Squeeze-and-Excitation (scSE) attention blocks into the decoder pathways.
   - Analyzed the model's improved capacity to recalibrate feature maps, effectively prioritizing relevant spatial features of small building footprints and intelligently suppressing irrelevant background noise.
   - Began preliminary integration of DeepLabV3+ with Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale context, testing its efficacy on varied building scales.

Work Done by Member C - Track C

1. **Focal Tversky Loss Experimentation:**
   - Substituted the baseline CE+Dice loss function with the Focal Tversky Loss to explicitly and aggressively penalize false negatives. This is a crucial adjustment for severely underrepresented minority classes where missing a destroyed building (false negative) is significantly more detrimental to disaster response than predicting one erroneously (false positive).
   - Conducted independent 40-epoch trials with varying hyperparameters to meticulously tune the false positive/false negative trade-off. Trial 1 utilized $\alpha = 0.3, \beta = 0.7, \gamma = 2.0$ to heavily bias the network toward recall, while Trial 2 tested a more balanced approach with $\alpha = 0.25, \beta = 0.75, \gamma = 1.33$ for a smoother gradient curve during backpropagation.
   - Evaluated the loss function's ability to force the network to learn optimal structural boundaries for the 'Destroyed' class. The non-linear nature of the focusing parameter (γ) proved essential in preventing the dominant background class from overwhelming the gradient updates.
2. **Phased Compound Loss Scheduler Development:**
   - Developed a dynamic, epoch-aware custom loss module (`PhasedCompoundLoss`) specifically designed to optimize the training progression, prevent early divergence, and maximize the final evaluation metrics.
   - Configured the training loop to exclusively utilize Focal Tversky loss for the first 80% of epochs. This initial phase is strictly dedicated to rapidly stabilizing minority class gradients and establishing robust basic feature extraction across all convolutional layers.
   - Programmed an automated epoch-trigger to inject Lovász-Softmax loss—a direct continuous and differentiable surrogate for the discrete Jaccard index (IoU)—for the final 20% of training. This strategic two-stage approach effectively fine-tunes and smooths the final segmentation boundaries directly for our evaluation metric without destabilizing the network early on.

Work Done by Member D - Track D

1. **Resolution and Cropping Ablation Pipeline:**
   - Evaluated the direct impact of training resolution strategies using the Albumentations library, holding all architectural, data sampling, and loss variables strictly constant to ensure pristine and undeniable ablation results.
   - Benchmarked the standard downsampling approach (Resize $1024 \rightarrow 512$) against native resolution random crops (RandomCrop 512×512). Empirical results definitively proved that preserving native pixel scale is critical for detecting localized structural damage, as downsampling irreversibly destroys high-frequency textural details.
   - Engineered a high-resolution pipeline for expanded crops (768×768). To accommodate this massive memory footprint, we utilized PyTorch gradient accumulation steps (yielding an effective Batch Size of 8). This allowed us to successfully train larger contextual patches, providing the network with a wider receptive field without triggering VRAM exhaustion (OOM errors) on Kaggle T4 GPUs.
2. **Test-Time Augmentation (TTA) and Sliding Window Inference:**
   - Developed a robust inference-time augmentation pipeline applying 8 distinct geometric transformations (horizontal/vertical flips, and 90, 180, 270 degree rotations) dynamically to each evaluation image batch.
   - Implemented complex tensor logic to automatically calculate the strict inverse transformations of the spatial predictions. By aligning and averaging the resulting softmax output maps, we significantly enhanced predictive reliability, smoothed boundary predictions, and mitigated hallucinatory artifacts.
   - Architected a sliding window inference function with spatial overlap stitching (e.g., 512×512 crop with a 128-pixel overlap). This function allows models trained strictly on small cropped patches to evaluate full-resolution 1024×1024 images seamlessly. To prevent grid-like artifacts at the crop boundaries, we integrated a spatial weighting scheme that prioritizes the center pixels of each crop.

Next Week's Strategic Roadmap

- **Phase 1 Completion:** We will execute the final set of isolated ablation experiments. This includes developing Copy-Paste instance augmentation pipelines (Member A), full DeepLabV3+ and MA-Net comparative evaluations (Member B), and advanced backbone swaps comparing ResNet50 against EfficientNet-B4 (Member C).
- **Champion Model Integration:** Upon completion of all isolated experiments, we will scientifically merge the absolute most successful variables from each member's respective track into a unified "Champion Model" and initiate a final training run to achieve our maximum theoretical mIoU.
- **Contingency Phase 2 Activation:** Should the Champion Model fail to consistently breach the target 0.52 mIoU threshold, we will formally initiate Phase 2. This involves conditionally triggering external dataset extensions, specifically surveying and converting external disaster data to forcefully supplement our minority class pools.